

2.17

Oyster Shell

2.1% A

55% B

42.9% I

M₁ limo

ABEF

M₂ clay

ACDI

M₃ Mill

100% D

M₄ oyster

ABI

M₅ 100% I (gas)M₆ = 100 tons/day

21% A

65% B

1-54% D

X_{6C} M₆ = CX_{6E} M₆ = E

Kevin Malec

84/84

0 = in - out + gen - con
Species balance

$$A: .01m_1 + .68m_2 + 0m_3 + .021m_4 = .21(100 \text{ tons})$$

$$B: .54m_1 + 0m_2 + 0m_3 + .55m_4 = .65(100 \text{ tons})$$

$$C: 0 + .2m_2 + 0 + 0 = w_{6C}(100 \text{ tons})$$

$$D: 0 + .04m_2 + m_3 + 0 = w_{6D}(100 \text{ tons})$$

$$I: .41m_1 + .09m_2 + 0 + .429m_4 = m_5$$

$$\text{total nonI: } .54m_1 + .92m_2 + m_3 + .571m_4 = 100 \text{ tons}$$

$$\text{total: } m_1 + m_2 + m_3 + m_4 = m_5 + 100 \text{ tons}$$

$$E: .04m_1 + 0 + 0 + 0 = w_{6E}$$

	m ₁	m ₂	m ₃	m ₄	m ₅	w _{6E}	w _{6C}
A	✓	✓		✓			
B	✓			✓			
C		✓					✓
D		✓	✓				
I	✓	✓			✓		
total nonI	✓	✓	✓	✓			
E	✓					✓	

I can't find a way to solve for any variable

$$\text{cost} = \left[(82.4)(80) + (28.5)(35) + (37.3)(60) \right] \frac{\text{tons}}{\text{day}} \frac{\$}{\text{ton}} = \$10486.70 \text{ per day}$$

2.167

m1	m2	m3	m4	m5	wc	we	B
0.01	0.68	0	0.021	0	0	0	21
0.54	0	0	0.55	0	0	0	65
0	0.2	0	0	0	-100	0	0
0	0.04	1	0	0	0	0	5
0.04	0	0	0	0	0	-100	0
0.41	0.08	0	0.429	-1	0	0	0
0.59	0.92	1	0.571	0	0	0	100

inverse A

-30.0099	-22.9291	0	-23.1895	0	0	23.18948
1.001984	-0.41419	0	-0.3621	0	0	0.362103
-0.04008	0.016567	0	1.014484	0	0	-0.01448
29.46429	24.33036	0	22.76786	0	0	-22.7679
0.41627	1.003671	0	0.230754	0	-1	-0.23075
0.002004	-0.00083	-0.01	-0.00072	0	0	0.000724
-0.012	-0.00917	0	-0.00928	-0.01	0	0.009276

m1	82.40327 tons/day
m2	28.51935 tons/day
m3	3.859226 tons/day
m4	37.27679 tons/day
m5	52.05863 tons/day
Wc	0.057039
We	0.032961

21

$$\frac{dm_c}{dt} = -.116 m_c$$

m_c in mg
 t in hr

see attached graphs

$$\int_{100}^{200} \frac{dm_c}{m_c} = \int_0^t -.116 dt \quad \ln\left(\frac{1}{100}\right) - \ln\left(\frac{1}{200}\right) = -.116 t_f$$

200

$$\frac{\ln(100) - \ln(200)}{-.116} = t_f = 5.98 \text{ hr to drop from 200mg to 100mg}$$

at 11 PM after 1 cup of coffee, you have 27.8 mg caffeine

at 11 PM after 1 cup at 6 and 1 cup at 2 PM, you have 98.2 mg caffeine

It is a significant increase in caffeine. If you are having trouble sleeping, you should either not drink the second cup or drink a smaller second cup

3.24

CH₃OH
100 kgmol/h
O₂
21.05 kgmol/h



HCOOH
CO
CO₂
H₂O
O₂

HCHO

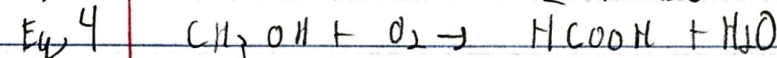
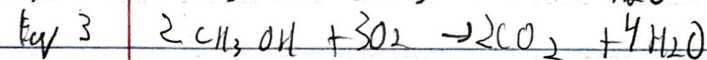
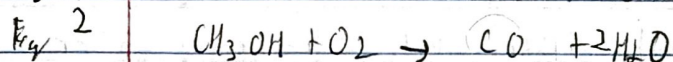
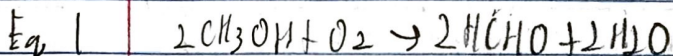
40% methanol reacts

95% O₂ reactsCH₃OH60% methanol doesn't react $\Rightarrow$ CH₃OH out = .6 CH₃OH in

$$100 \frac{\text{kgmol}}{\text{h}} \times .6 = 60 \frac{\text{kgmol}}{\text{h}} \text{ CH}_3\text{OH}$$

40 kgmol reacts/h

$$5\% \text{ O}_2 \text{ doesn't react} \Rightarrow .05 \times 21.05 \frac{\text{kgmol}}{\text{h}} = 1.05 \frac{\text{kgmol}}{\text{h}} \text{ O}_2$$

1.05 kgmol O₂/h

20 kgmol reacts/h

$$\text{O}_2 \text{ bal } 0 = -21.05 \frac{\text{kgmol}}{\text{h}} - 2\dot{\epsilon}_1 - \dot{\epsilon}_2 - 3\dot{\epsilon}_3 - \dot{\epsilon}_4 + 1.05 \frac{\text{kgmol}}{\text{h}}$$

$$\text{CH}_3\text{OH bal } 0 = -100 \frac{\text{kgmol}}{\text{h}} - 2\dot{\epsilon}_1 - \dot{\epsilon}_2 - 2\dot{\epsilon}_3 - \dot{\epsilon}_4 + 60 \frac{\text{kgmol}}{\text{h}}$$

$$0 = 78.95 + \dot{\epsilon}_1 + 0 - \dot{\epsilon}_3 - 0 = 58.95$$

$$20 = \dot{\epsilon}_1 - \dot{\epsilon}_3 \quad \dot{\epsilon}_3 \text{ cannot be negative}$$

$$\Rightarrow 20 = \dot{\epsilon}_1, \quad \dot{\epsilon}_2, \dot{\epsilon}_3, \dot{\epsilon}_4 = 0$$

$\Rightarrow$ Eq 2, 3, 4 do not happen $\Rightarrow$ no CO, CO₂, HCOOH produced

$$\text{HCHO balance } 0 = 0 + \text{Fout} + 2\dot{\epsilon}_1$$

$$\text{Fout} = -2\dot{\epsilon}_1 = -2 \times 20 = 40 \frac{\text{kgmol}}{\text{h}} \text{ HCHO}$$



3.21

top graph

1 cup of coffee

$$y = 279e^{(-.116t)}$$

bottom graph

caffeine after 2nd cup at 2 pm

caffeine with 1 cup at $t = 0$ and 1 cup at $t = 8$

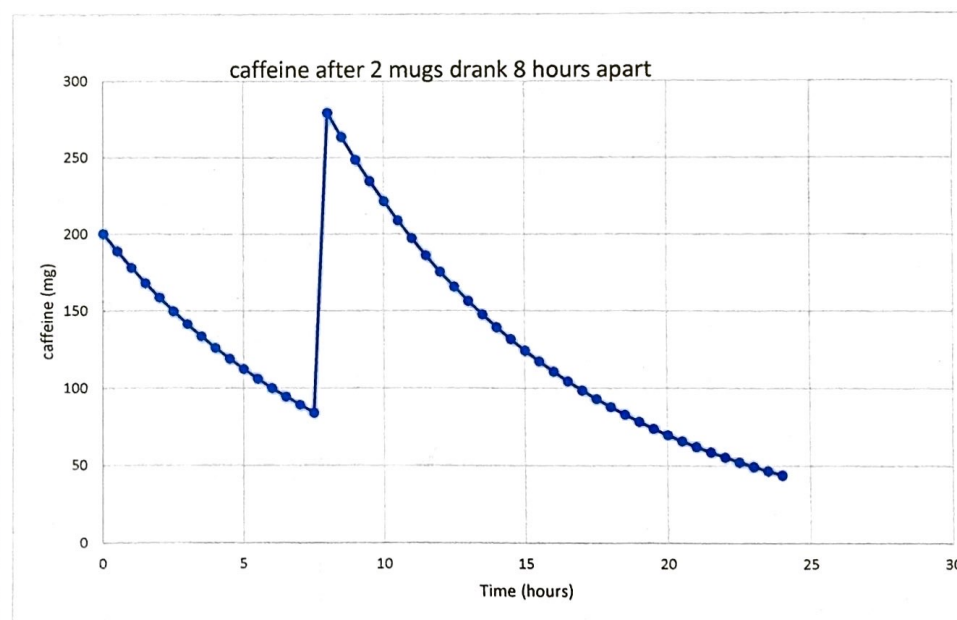
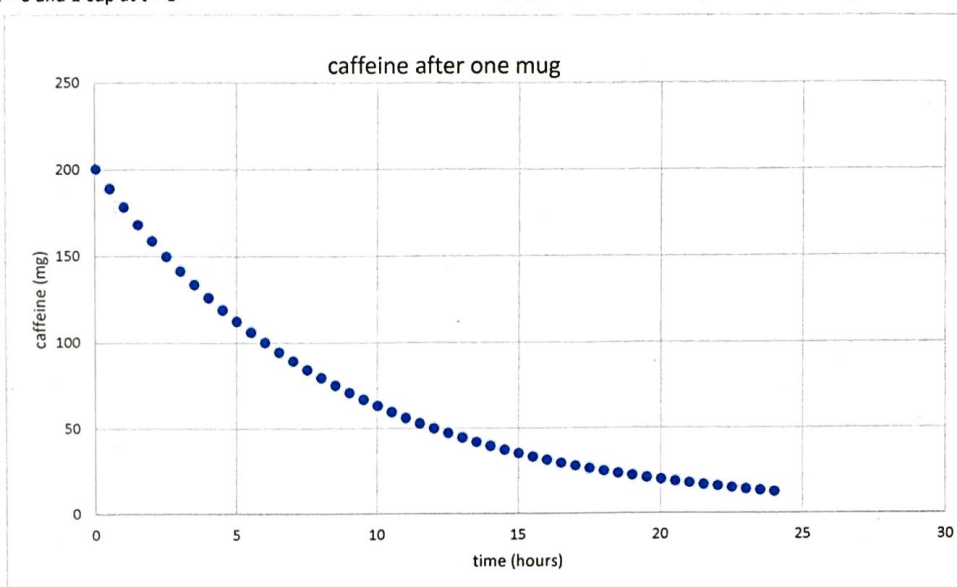
$y = 200e^{(-.116t)}$

0	200
0.5	188.73
1	178.095
1.5	168.0594
2	158.5892
2.5	149.6527
3	141.2198
3.5	133.262
4	125.7527
4.5	118.6665
5	111.9797
5.5	105.6696
6	99.71512
6.5	94.09617
7	88.79385
7.5	83.79031
8	79.06872
8.5	74.61319
9	70.40874
9.5	66.4412
10	62.69724
10.5	59.16424
11	55.83034
11.5	52.68429
12	49.71553
12.5	46.91406
13	44.27045
13.5	41.77581
14	39.42174
14.5	37.20032
15	35.10408
15.5	33.12596
16	31.25931
16.5	29.49785
17	27.83564
17.5	26.2671
18	24.78695
18.5	23.39021
19	22.07217
19.5	20.8284
20	19.65472
20.5	18.54717
21	17.50204
21.5	16.5158
22	15.58513
22.5	14.70691
23	13.87817
23.5	13.09614
24	12.35817

t=17 is 11pm

0	279.0687
0.5	263.3432
1	248.5038
1.5	234.5006
2	221.2865
2.5	208.817
3	197.0501
3.5	185.9463
4	175.4682
4.5	165.5806
5	156.2501
5.5	147.4454
6	139.1369
6.5	131.2965
7	123.8979
7.5	116.9163
8	110.328
8.5	104.111
9	98.24438
9.5	92.70831
10	87.48419
10.5	82.55445
11	77.9025
11.5	73.51269
12	69.37025
12.5	65.46123
13	61.77249
13.5	58.2916
14	55.00687
14.5	51.90723
15	48.98225
15.5	46.2221
16	43.61748

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enters at $\frac{62000 \text{ molecules}}{\text{hr}}$ leaves at $\frac{57000 \text{ molecules}}{\text{hr}}$

$$\text{degradation } 2700(1 - e^{-.3t})$$

must accumulate 30,000 molecules in 8 hours

accumulation $\approx$ in - out + gen - con degradation $\approx$ consumed

$$\approx 62000 - 57000 + 0 - 2700(1 - e^{-.3t})$$

$$\text{accum} = \int_0^8 62000 dt - \int_0^8 57000 dt - \int_0^8 2700(1 - e^{-.3t}) dt$$

$$\text{accum} = 496000 - 456000 - 2700 \int_0^8 dx - 2700 \int_0^8 e^{-.3x} dx$$

$$\approx 40000 - 13416.5 - 1100 e^{-.3x}$$

$$= \boxed{26582.5 \text{ molecules}} < 30,000 \text{ molecules}$$

NO, do not invest